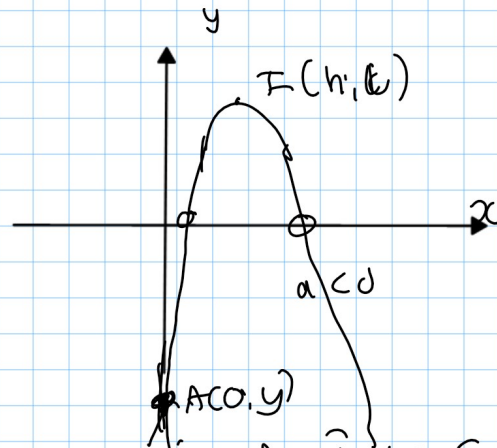
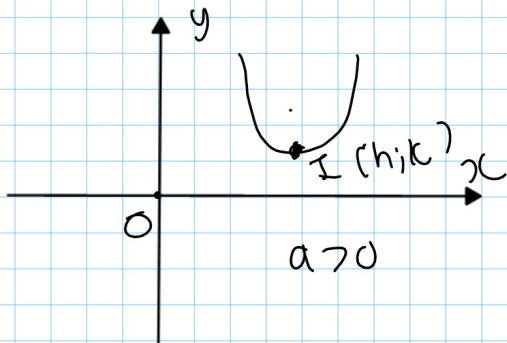


Part 2: Advanced maths

Parabola, quadratic function (PT bài 2)
 Exponential (hàm mũ)
 Radicals (căn bậc 2, 3)
 Rational exponent (số mũ phân)

1. Quadratic function
 $y = ax^2 + bx + c$

1. Critical value



- Vertex (đỉnh) của parabola: Khi $a > 0$, vertex là min; Khi $a < 0$, vertex là max.
- y-intercept (y-axis): initial value (giá trị ban đầu), $A(0, y)$.
- x-intercept (x-axis): Solution ($y = 0$).

2. Forms:

$$y = ax^2 + bx + c$$

$A(x_1, 0)$
 $B(x_2, 0)$ } là 2 nghiệm của pt

- Intercept forms: $y = a(x - x_1)(x - x_2)$
 a : coefficient

$I(h, k)$

- Vertex forms: $y = a(x - h)^2 + k$, với h, k là tọa độ, vô đỉnh parabola

3. Quadratic formula

$$y = ax^2 + bx + c$$

$$\Delta = b^2 - 4ac$$

$$I\left(\frac{-b}{2a}, \frac{-\Delta}{4a}\right)$$

$$y = 1x^2 - 4x + 3$$

$$a = 1$$

$$b = -4$$

$$c = 3$$

$$\Delta = 4$$

$$I(2, -1)$$

TH1: $\Delta > 0$ thì có 2 nghiệm phân biệt

$$x_1 = \frac{-b + \sqrt{\Delta}}{2a}$$

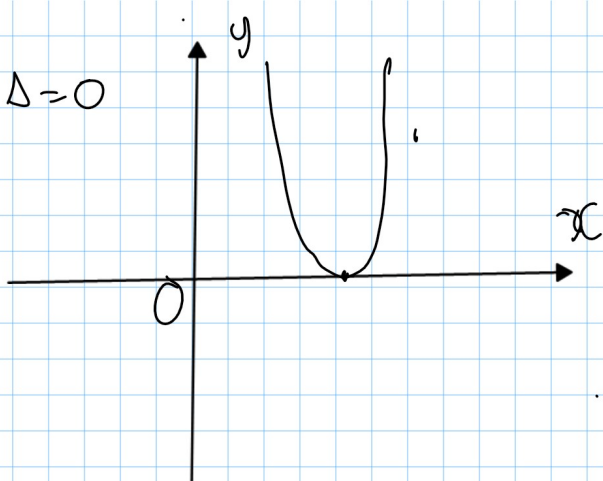
$$x_2 = \frac{-b - \sqrt{\Delta}}{2a}$$

TH2: $\Delta = 0$ có nghiệm kép

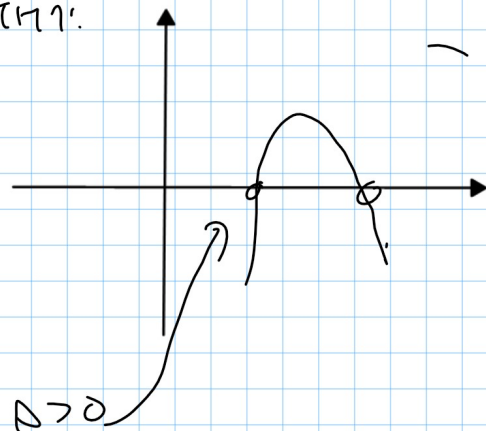
$$x_1 = x_2 = \frac{-b}{2a}$$

TH3: $\Delta < 0$ là vô nghiệm

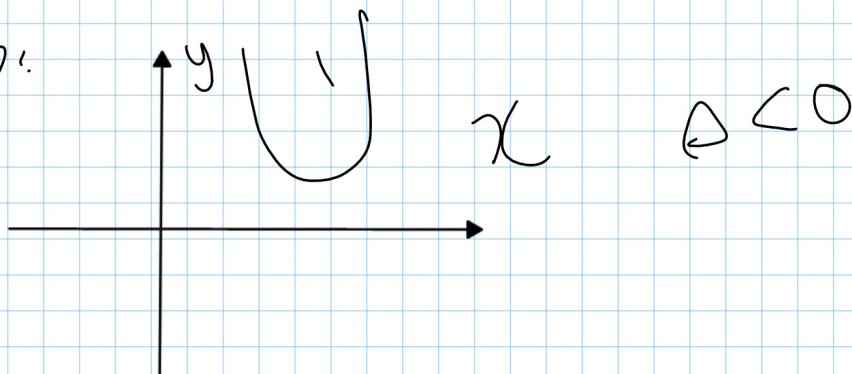
TH2: $\Delta = 0$



TH1:



TH3:



4. Định lý Viète (Viète theorem)

$y = ax^2 + bx + c$ for 2 nghiệm x_1, x_2

$$x_1 x_2 = \frac{c}{a}$$

product of x_1, x_2

Quotient: Chia (Thương)
Subtraction: (-) (Hiệu)

$$x_1 + x_2 = -\frac{b}{a}$$

Sum of x_1, x_2

II. Exponential function (Hàm số mũ)

$$y = a \cdot b^x$$

a: initial number (giá trị ban đầu) $(0 < a)$
b: growth factor
x: total time for a to be multiplied by b

$$y = 2 \cdot (1.5)^x$$

III. Radicals (Căn số)

- Square roots: $\sqrt{a}$ (căn bậc 2)
- Cube roots: $\sqrt[3]{a}$ (căn bậc 3)
- root: Căn

IV. Rational exponents (Số hữu tỉ)

- Conversion:

$$\sqrt{2} = 2^{1/2}$$

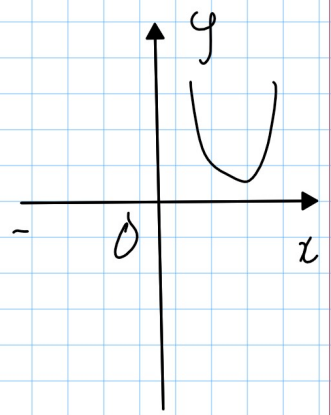
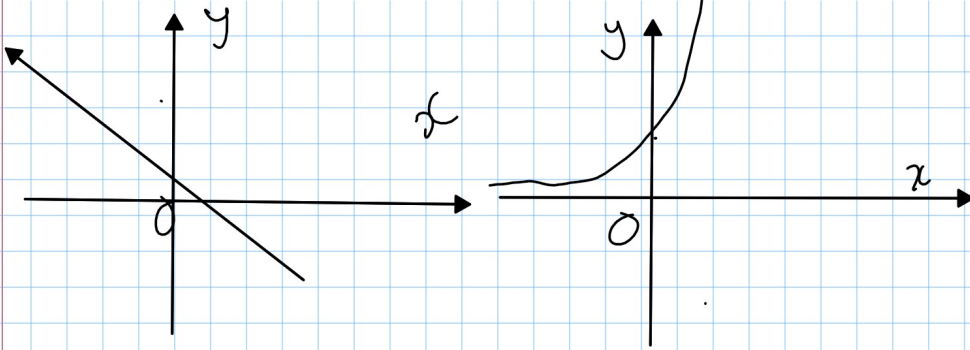
$$\sqrt[3]{2} = 2^{1/3}$$

V. Phân biệt (Things to distinguish)

linear
f(x)

exponential
g(x)

Quadratic
h(x)



$$\left. \begin{array}{l} (x_1, y_1) \\ (x_2, y_2) \\ (x_3, y_3) \end{array} \right\} \in f(x), g(x)$$

linear: $y_2 - y_1 = y_3 - y_2$

exponential: $\frac{y_2}{y_1} = \frac{y_3}{y_2}$